

## Integrable Systems — Exercise Sheet 14

The quantum inverse scattering method

Variant: Questions only

Conventions, as in Lecture 14:  $\hbar = 1$  except in Exercises 3 (iii), 5 and 6, where  $\hbar > 0$ . The auxiliary spaces  $a, b$  and the sites  $n = 1, \dots, N$  are copies of  $\mathbb{C}^2$ , and  $P_{xy}$  exchanges the factors  $x$  and  $y$ .  $R(u) = u - iP$ ; the Lax operator of the Heisenberg chain is  $L_{an}(\lambda) = \lambda + iP_{an}$ ; the monodromy is  $T_a(\lambda) = L_{aN}(\lambda) \cdots L_{a1}(\lambda)$  and the transfer matrix  $t(\lambda) = \text{tr}_a T_a(\lambda)$ . The  $RTT$  relation is  $R_{ab}(\lambda - \mu)T_b(\mu)T_a(\lambda) = T_a(\lambda)T_b(\mu)R_{ab}(\lambda - \mu)$ , and  $C = P_{ab}P_{an} = P_{an}P_{bn} = P_{bn}P_{ab}$ ,  $C^{-1} = P_{an}P_{ab} = P_{ab}P_{bn} = P_{bn}P_{an}$ . Classically,  $\{f, g\} = \sum_n (\partial_{q_n} f \partial_{p_n} g - \partial_{p_n} f \partial_{q_n} g)$ , and legs are as in Lecture 9:  $T_1 = T \otimes \mathbb{1}$ ,  $T_2 = \mathbb{1} \otimes T$ ,  $(A \otimes B)_{(ac),(bd)} = A_{ab}B_{cd}$ .

**Exercise 1 (the Lax operator of the Heisenberg chain).** (i) On  $\mathbb{C}_a^2 \otimes \mathbb{C}_n^2$  show that  $P_{an} = \frac{1}{2}(\mathbb{1} + \sigma_a \cdot \sigma_n)$ . (ii) Deduce Lecture 14's matrix form of  $L_{an}(\lambda)$  in the auxiliary space, with entries  $\lambda + \frac{i}{2} \pm iS_n^\pm$  on the diagonal and  $iS_n^\pm$  off it. (iii) Write  $L_{an}(\lambda)$  as a  $4 \times 4$  matrix in the basis  $|\uparrow\uparrow\rangle, |\uparrow\downarrow\rangle, |\downarrow\uparrow\rangle, |\downarrow\downarrow\rangle$  (auxiliary state first). Check that exactly six entries are not identically zero (as functions of  $\lambda$ ), and explain, with the vertex picture of Lecture 14, why two of them are  $\lambda + i$  and not  $\lambda$ . (iv) For which  $\lambda$  is  $L_{an}(\lambda)$  not invertible? Show that there it is a multiple of a projector, and that  $L_{an}(0) = iP_{an}$ .

**Exercise 2 (RLL fixes  $R$ ).** Let  $R_{ab} = \alpha + \beta P_{ab}$  with numbers  $\alpha, \beta$  (which may depend on  $\lambda, \mu$ ). (i) Following the proof of Lecture 14's  $RLL$  theorem, show that

$$R_{ab}L_{bn}(\mu)L_{an}(\lambda) - L_{an}(\lambda)L_{bn}(\mu)R_{ab} = [\alpha - i\beta(\lambda - \mu)](C - C^{-1}).$$

(ii) Deduce that the non-zero  $R_{ab}$  of this form satisfying  $RLL$  are exactly the non-zero multiples of  $R_{ab}(\lambda - \mu) = \lambda - \mu - iP_{ab}$ . (iii) At  $\lambda = \mu$  this  $R$  is a multiple of  $P_{ab}$ . Show directly that  $P_{ab}L_{bn}(\lambda)L_{an}(\lambda) = L_{an}(\lambda)L_{bn}(\lambda)P_{ab}$ .

**Exercise 3 (the order of  $RTT$ ).** Let  $T$  satisfy  $RTT$  with  $R(u) = u - iP$ . (i) Using  $R(u)R(-u) = -(u^2 + 1)\mathbb{1}$  (Lecture 14), show that for  $\lambda - \mu \neq \pm i$  the  $RTT$  relation is equivalent to  $\tilde{R}_{ab}(\lambda - \mu)T_a(\lambda)T_b(\mu) = T_b(\mu)T_a(\lambda)\tilde{R}_{ab}(\lambda - \mu)$ , with  $\tilde{R}(u) = u + iP$ : the other order of the monodromies, with the other sign in  $R$ . (ii) Show that  $T'(\lambda) = T(-\lambda)$  satisfies the other order with the same  $R$ :  $R_{ab}(\lambda - \mu)T'_a(\lambda)T'_b(\mu) = T'_b(\mu)T'_a(\lambda)R_{ab}(\lambda - \mu)$  for all  $\lambda, \mu$  (where  $T$  is a polynomial in  $\lambda$ , as for the chain). For the Heisenberg chain, what is the Lax operator of  $T'$ ? (iii) Classically, let  $T$  satisfy Lecture 9's  $\{T_1(\lambda), T_2(\mu)\} = [r(\lambda - \mu), T_1(\lambda)T_2(\mu)]$  with  $r(\lambda) = -P/\lambda$ . Show that  $T'(\lambda) = T(-\lambda)$  satisfies the same bracket with  $-r$  in place of  $r$ . Then, restoring  $\hbar$  ( $R = \lambda - i\hbar P = \lambda(\mathbb{1} + i\hbar r)$ ), expand the relation of (ii) to order  $\hbar$  as in Lecture 14's classical limit, taking the symbol of  $T'_a T'_b - T'_b T'_a$  to be  $i\hbar \{T'_1, T'_2\} + O(\hbar^2)$ , and check that it gives the same.

**Exercise 4 (three sites by hand).** Take the Heisenberg chain with  $N = 3$ . (i) Show that  $\text{tr}_a P_{an} = \mathbb{1}$ ,  $\text{tr}_a(P_{an}P_{am}) = P_{nm}$  for  $n \neq m$ , and  $\text{tr}_a(P_{a3}P_{a2}P_{a1}) = P_{32}P_{31}$ , and deduce

$$t(\lambda) = 2\lambda^3 + 3i\lambda^2 - \lambda(P_{12} + P_{13} + P_{23}) - iP_{32}P_{31}.$$

(ii) Show that  $U = P_{32}P_{31}$  sends  $x_1 \otimes x_2 \otimes x_3$  to  $x_3 \otimes x_1 \otimes x_2$ , so that  $t(0) = -iU$  is  $-i$  times the shift by one site. Show that  $U$  commutes with  $P_{12} + P_{13} + P_{23}$ , and deduce  $[t(\lambda), t(\mu)] = 0$

directly. (iii) Show that  $i t(0)^{-1} t'(0) = P_{12} + P_{23} + P_{31}$ , and so that Lecture 10's Hamiltonian at  $N = 3$  is  $H = \frac{j}{2} (i t(0)^{-1} t'(0) - 3)$ . (iv) Find the eigenvalues of  $t(\lambda)$  and of  $H$ , with their multiplicities.

**Exercise 5 (symbols).** One degree of freedom,  $[q, p] = i\hbar$ ,  $p = -i\hbar \partial_q$  on functions of  $q$ . (i) Show that  $p g(q) = g p - i\hbar g'$ , and by induction  $p^k g = \sum_{j=0}^k \binom{k}{j} (-i\hbar)^j g^{(j)} p^{k-j}$ : the symbol formula of Lecture 14 for  $F = p^k$ ,  $G = g$ . (ii) Compute  $[p^2, e^q]$  exactly, and compare its symbol with  $i\hbar \{p^2, e^q\}$ . (iii) Show that  $[p, g(q)] = i\hbar \{p, g\}$  exactly, for every  $g$ . Deduce that the commutators of the entries of the quantum Toda site matrix  $\begin{pmatrix} \lambda - p & e^{-q} \\ -e^q & 0 \end{pmatrix}$  are exactly  $i\hbar$  times the Poisson brackets of the corresponding functions. Why does (ii) not contradict this?

**Exercise 6 (the quantum Toda chain,  $N = 2$  and 3).** Quantise Lecture 9's periodic Toda chain as in Lecture 14:  $[q_m, p_n] = i\hbar \delta_{mn}$ ,  $L_n(\lambda) = \begin{pmatrix} \lambda - p_n & e^{-q_n} \\ -e^{q_n} & 0 \end{pmatrix}$ ,  $t(\lambda) = \text{tr}(L_N(\lambda) \cdots L_1(\lambda)) = \sum_k c_k \lambda^{N-k}$ . (i) For  $N = 2$ , find  $c_1$  and  $c_2$ , and show that  $H = \frac{1}{2} c_1^2 - c_2 = \frac{1}{2} (p_1^2 + p_2^2) + e^{q_1 - q_2} + e^{q_2 - q_1}$ . (ii) Show directly that  $[c_1, c_2] = 0$  for  $N = 2$ . (iii) For  $N = 3$ , show that  $c_3 = -p_1 p_2 p_3 + p_1 e^{q_2 - q_3} + p_2 e^{q_3 - q_1} + p_3 e^{q_1 - q_2}$ , and explain why no ordering question arises. Then verify  $[H, c_3] = 0$  by hand, with  $H = \frac{1}{2} \sum_n p_n^2 + e^{q_1 - q_2} + e^{q_2 - q_3} + e^{q_3 - q_1}$ , and check that the terms of order  $\hbar^2$  cancel too.

**Exercise 7 ★. Any spin.** Let  $\mathbf{S}$  be spin operators on a space  $V$ ,  $[S^j, S^k] = i\epsilon_{jkl} S^l$ , and on  $\mathbb{C}_\alpha^2 \otimes V$  put  $L_{an}(\lambda) = \lambda + \frac{1}{2} + i\sigma_\alpha \cdot \mathbf{S}$ . Lecture 14 cites that it satisfies  $RLL$  with the same  $R$ ; prove it. (i) With  $X = \sigma_a \cdot \mathbf{S}$  and  $Y = \sigma_b \cdot \mathbf{S}$ , show that  $RLL$  for all  $\lambda, \mu$  is equivalent to  $[X, Y] = (Y - X)P_{ab}$ . (ii) Prove this identity, using only the commutation relations of  $\mathbf{S}$ . (iii) For an irreducible spin  $s \geq \frac{1}{2}$ , find the eigenvalues of  $L_{an}(\lambda)$  and their multiplicities, and show that for  $s = \frac{1}{2}$  you recover  $\lambda + iP_{an}$ .